QR-009 Words the team uses

Doc ID	QR-009
Title	Words the team uses
Parent	CS-001 §4; CS-002 §4; CS-003 §4; CS-004 §4; CS-005 §4; CS-006 §4; CS-007 §4; CS-008 §4; CS-009 §4; CS-011 §4; CS-012 §4; CS-013 §4; CS-013 §4.1; CS-014 §4
Checked against	CS-001 Rev G; CS-002 Rev F; CS-003 Rev E; CS-004 Rev E; CS-005 Rev F; CS-006 Rev D; CS-007 Rev D; CS-008 Rev E; CS-009 Rev D; CS-011 Rev E; CS-012 Rev G; CS-013 Rev G; CS-013 Rev G; CS-014 Rev A

Derived from the Definitions sections of its parent standards. If this and a parent disagree, the parent is right.

Each section is one family of words. The aliases are the point: if you heard a word at RFS and do not know it, search for it and the section that lists it explains it.

1. Car and part names: what do UT, RW and FW mean?

UT is the undertray, RW the rear wing and FW the front wing. SN5 is the team's fifth car and SN6 the sixth; the season code is the middle of every ID, as in `P-SN6-007`, and SN5 IDs stay frozen (CS-013 §4.1).

Also called: UT, undertray, under tray, floor; RW, rear wing, wing; FW, front wing; SN5, SN6, the car; aero parts.

- Undertray (UT): the inlet, the diffuser, the left and right side pods, the strakes and the air dams.
- Rear wing (RW): the elements and the endplates. The RW E2 to E5 elements are the foam-wrapped class of part (CS-007 §7.7).
- Front wing (FW).
- Other composites parts: nosecone, clamshell, dashboard, seat, steering wheel, catch can.
- Catch can: the SN5 part with carbon over a thin metal can, the cautionary tale for sanding through and for grounding (CS-009 §1, CS-010 §7.1).
- Forged CF: compression-molded carbon, made in 3D-printed molds (CS-003 Appendix A).
- Aero parts are often defined in #aerodynamics; composites builds them.

2. Mold and tooling board words: flange, scribe line, mold file, stage

A mold is cut on the ShopSabre from tooling board glued into a stack. The flange is the flat margin round the cavity for tacky tape and lines, and scribe lines are machined V-grooves marking the trim line and datums (CS-003 §4).

Also called: mold, mould, tool; tooling board, board, Precision Board, PBLT; stack, block, mold stock, blank; flange; scribe line, etch line, trim line.

- Stack (mold stock): tooling boards glued into a block thick enough for the mold. Not the same as a layup stack (QR-009 §10).
- Mold record: `MOLD-SN6-###`, on the Molds tab, from the moment the stack is planned.
- Mold file: the mold's current stack plan, `STK-SN6-###`. Older plans show under Earlier, marked Superseded.
- Mold stage: Designed, Tooling cut, Board glued, Machined, Sealed, Ready for layup; Retired sits off the track. A mold that will not be used again is retired, never deleted.
- Datum cut plan: the drawing of reference cuts taken off a mold after a part comes out. Most molds never need one.

- Sealing: closing the open cells of machined board so resin cannot soak in and lock the part on. XCR does it (CS-004 §4).
- Release system: wax, then PVA film, between the sealed mold and the part.
- B-stage, firm-but-tacky: resin firm enough not to flow but soft enough to dent with a fingernail; the right time to overcoat (CS-004 §4).

3. Bagging consumables: peel ply, breather, flow mesh, tacky tape

Peel ply is the release fabric on the laminate, breather the lofted cloth under the bag, flow mesh the green mesh that spreads resin in an infusion, and tacky tape seals the bag to the flange (CS-006 §5, CS-007 §4).

Also called: consumables; peel ply, release ply, PP180; breather, bleeder, batting; flow mesh, infusion mesh, flow media, green mesh, 2-in-1 mesh; tacky tape, sealant tape, AT-200Y, yellow tape; bag, vacuum bag, VB160, bag film; spiral, spiral wrap; resin trap, catch pot.

- Bag film: VB160 (CS-006 §5).
- Peel ply: PP180, or Sigmatex-supplied (CS-006 §5).
- Breather: gives air a path to the vacuum port and soaks up excess resin. Kept clear of part edges, where it glues on (CS-007 §4).
- Flow mesh: covers the whole part in an infusion. The 2-in-1 mesh is mesh and peel ply in one (CS-006 §5).
- Tacky tape: Airtech AT-200Y (yellow) sealant tape (CS-006 §5).
- Spiral wrap and PE tubing: the feed and vacuum lines (CS-006 §5).
- Resin trap, catch pot: the sealed pot between the bag outlet and the pump that catches over-pulled resin. Without it the first overlong pull kills the pump (CS-006 §4).
- Pleats: folds built into the bag with tacky tape so the film reaches into corners (CS-006 §4).
- Bridging: bag film spanning a corner instead of following it. It punctures or leaves a resin-rich void (CS-007 §4).

4. Infusion and wet layup words: drop test, inHg, wet-out, race-tracking

A drop test (leak-down) shuts the bag off from the pump and watches the gauge, to find leaks before resin goes in. inHg, inches of mercury, is what the team's vacuum gauges read; full vacuum at sea level is about 29.9 inHg (CS-006 §4).

Also called: drop test, leak test, leak-down, leak check; inHg, inches of mercury, vacuum, vac; infusion, infuse, VARTM, resin infusion; wet layup, wet lay, hand layup; wet-out, wetting out; race-tracking, racing.

- Nothing reads "30 psi of vacuum": atmospheric pressure is only about 14.7 psi. A gauge showing psi is reading something else (CS-006 §4).
- Race-tracking: resin finding a fast path, usually along an edge, and reaching the outlet before the middle is wet (CS-006 §4).
- Wet-out: cloth fully saturated, a uniform sheen with the weave still visible. Swimming in resin is past wet-out, not better (CS-007 §4).
- 60:40: the target cured fiber to resin ratio, by weight (CS-007 §4).
- Infusion: the dry stack is bagged and resin is pulled through it under vacuum (CS-006). Wet layup: plies are wetted by hand, then bagged (CS-007).
- Foam wrapped: wet plies wrapped round a shaped foam core instead of laid in a mold (CS-007 §7.7).
- Double-pot: mix, pour into a second cup, mix again with a fresh stick (CS-008 §7 step 3).
- Degas, degassing: an optional step that pulls the air out of mixed IN2 under vacuum before it is infused (CS-006 §7.8).

5. Resin names: IN2, AT30, West, XCR, DP420

IN2 with AT30 hardener is the infusion resin, West System 105 with 205, 206 or 209 is the wet layup resin, XCR seals molds, and DP420 is the structural adhesive. Ratios and holds per system are in CS-008 §5 and

QR-001.

Also called: resin, epoxy, hardener, activator, Part B; IN2, infusion resin; AT30, AT30 SLOW, AT30 FAST; West, West System, 105, 205 fast, 206 slow, 209 extra slow; XCR, sealer, coating resin; DP420, 3M glue, structural adhesive; 404, filler, thickened epoxy; clear coat, LiME LiNE.

- IN2 and AT30 come from Easy Composites (EC). AT30 comes in SLOW and FAST (CS-008 §5).
- West 105 is the resin; 205, 206 and 209 are its hardeners. 206 is the wet layup default (CS-008 §5).
- 209 uses 3:1 pumps, not the 5:1 pumps of 205 and 206: the classic wrong-ratio trap (CS-008 §7 step 1).
- XCR: Easy Composites' epoxy coating resin for mold sealing (CS-004).
- DP420: 3M Scotch-Weld structural adhesive in a duo-pak cartridge (CS-009 §5).
- West 404: high-density filler for thickened epoxy, for fillets and gap filling (CS-009 §5).
- LiME LiNE UC4:1: the urethane clear coat (CS-009 §5).
- Ure-Bond 90: the tooling-board glue, a urethane (CS-003 §6).
- Airtac 2: the spray tack for holding carbon and mesh on a mold (CS-006 §5).

6. Cure words: pot life, gel, demould (demold), FEB hold, full cure, exotherm

The FEB hold is how long the team waits before demoulding, and it governs. It is longer than the datasheet figure on purpose and is a team decision, not a manufacturer claim (CS-008 §5.1). Pot life is the usable time in the cup (CS-008 §4).

Also called: demould, demold, de-mould, demoulding, demolding; FEB hold, cure hold, team hold, wait time; pot life, working time, open time; gel, gelled, kicked; full cure; exotherm, hot cup.

- Pot life: usable time in the mixing pot, at the mass and temperature the TDS states. A bigger batch or a warmer shop shortens it (CS-008 §4).
- Gel: the resin no longer flows. Gel is not cure; a gelled part pried off a mold drags resin with it (CS-008 §4).
- Demould: the earliest the part can leave the mold without damage (CS-008 §4).
- Datasheet cure figure: the manufacturer's number, at their temperature. FEB hold: the team's number, which the app enforces on the step after the cure starts (CS-008 §5.1).
- Full cure: full mechanical properties. Weeks, not hours, for every system (CS-008 §4).
- Exotherm: the cure's own heat. Mass in a cup speeds it up (CS-008 §4).
- Still curing: what the app says when you try to sign a held step early (CS-013 §4).
- Demould early?: the lead's way to sign through a hold, with a written reason (CS-008 §5.1).

7. What do the ID prefixes mean? WO, P, MOLD, RDS, CPN, STK

Every record ID reads `PREFIX-SNx-###`. WO is a work order, P a part, MOLD a mold, RDS an R&D study, CPN a coupon, STK a mold's stack plan (CS-013 §4.1). IDs are never reused.

Also called: ID, part number, P number, WO number, record ID, prefix, code.

Prefix	What it names
`WO`	a work order (the job)
`P`	a part
`MOLD`	a mold
`PNL`	a flat test panel
`BRD`	a tooling-board sheet or offcut
`FAB`	a fabric roll or usable offcut
`RSN`	a resin or hardener lot
`CON`	a consumable lot
`JIG`	a jig or fixture
`BIN`	a storage location

Prefix	What it names
RDS	an R&D study: a named set of coupons
CPN	one coupon, a single test piece in a study
PROJ	an issue (older records say ticket); no label
STK	a mold's stack plan, the Mold file; no label
BUY	a purchase; no label
DOC	a document on the Documents tab; no label

- An ID names a physical object, never a job; WO is the one exception (CS-013 §4.1 rule 1).
- An ID is capped at 14 characters by the QR code (CS-013 §4.1).
- Dates are fields, never part of an ID.
- A re-laid part keeps its P ID and gets a new run (CS-013 §4.1 rule 3; QR-009 §8).

8. Work order words: run, buy-off, blocker, cure hold, technique, issue

A run is one attempt at making a part; the app's Work Orders tab calls a run a work order (CS-013 §4). A buy-off is pressing a step's sign-off while signed in. A blocker step blocks everything after it until it is signed.

Also called: WO, work order, run, job, traveler, traveller; buy-off, sign-off, sign, signing; blocker, locked step, gate; cure hold, hold; technique, checklist, process, layup type; issue, ticket, , disposition; retro WO.

- Part and run: a part (P-SNx-###) is the durable thing the car needs, and it has many runs (CS-013 §4).
- Blockers: Stack frozen, Mold design review, the drop test, and the acceptance criterion of the Other technique (CS-013 §4).
- Evidence: what a step needs before it can be signed, such as a layup stack, the CAD, or a note (CS-013 §4).
- Training to sign: a step only someone holding the named training may sign. A lead may sign without it and must write why (CS-013 §4).
- Cure hold: the wait on the step after the one that starts a cure, as long as the FEB hold for the resin recorded (CS-013 §4).
- Technique: the step checklist a part's Layup type picks: Mold infusion, Glass infusion, Mold wet lay, Foam wrapped or Other (CS-013 §7.5).
- Issue: raised with Raise an issue or on a step; a run cannot complete until every issue has a Disposition, What happened and What was done (CS-013 §7.6 steps 1 and 3).
- OnHold: a run someone has parked. It blocks nothing (CS-013 §4).
- Retro WO: a back-filled SN5 record, WO-SN5-*; its unverifiable fields say "not recorded (retro)" (CS-013 §4).
- ME and RE: the Mold Engineer and the Manufacturing Engineer, the two engineer fields on a part (CS-013 §4; QR-009 §14).

9. Part, R&D part, study and coupon: which is which?

A part is something the car needs, built through runs. An R&D part is a real build that feeds a decision instead of the car, with every blocker, hold and evidence check still enforced. A study (RDS) holds coupons (CPN), test pieces with no work order and no cure hold (CS-013 §4).

Also called: R&D, R and D, research, test, trial; R&D part, R&D WO; study, R&D study, RDS; coupon, CPN, test piece, specimen, tensile coupon; test panel, PNL, flat panel; season part, deliverable.

- R&D builds: coupon campaigns, flat test panels, layup trials, sealing and grounding experiments, mold shakedowns (CS-013 §4).
- R&D is a property of the part, not a status. A work order inherits it from its part (CS-013 §4).
- An R&D part is kept off the Season tab, the Google Sheet mirror and the Parts and Work Orders lists. It lives on the R&D tab, and is marked R&D on screen, on the label and on the scan page (CS-013 §4).

- The R&D mark is a ruled box on the label's ID line. It is not a class word (CS-001 §4).
- Which to make: test pieces you will cut, measure and compare should go in a study. Anything with a mold, a checklist or a cure you need to sign should be a part and a run, marked R&D if it is not for the car (CS-013 §2).
- The panel a coupon is cut from is still a build; the pieces cut from it are bench records (CS-013 §2).

10. Stack and ply words: layup stack, ply table, stack frozen, as-built

The layup stack is the ordered list of plies from the mold surface (P1) outward, including cores, meshes and inserts (CS-002 §4). Stack frozen is the buy-off that locks a run's copy of it; the as-built is that copy as laid.

Also called: stack, layup, lay-up, layup schedule, ply table, ply book, plies; stack frozen, freeze, stack freeze; as-built; orientation, 0/90, ±45; 88, 195, 220, 670, spread tow, twill; sacrificial ply.

- Cloth naming: `<gsm> <weave>`, for example 88 spread-tow, 195 twill, 220 twill, 670 twill (Sigmatex). Always gsm, never "the thick one" (CS-002 §4).
- Orientation: warp direction against the part's 0° datum; 0/90 or ±45 for woven cloth (CS-002 §4).
- Inserts: a G10 insert is a ply in the table, like a core, with material and thickness (CS-002 §4).
- Plan and as-built: the plan is the stack on the part (Parts tab, Layup stack). The as-built is the copy each run takes when it starts. Editing the run's copy never changes the plan (CS-002 §4).
- Stack freeze: from the Stack frozen buy-off on, edits to the part's plan no longer reach that run (CS-002 §4).
- Sacrificial ply: the 88 spread-tow outer layer that absorbs sanding. It is the sanding budget; structural plies are not (CS-009 §4).

11. Material and inventory words: lot, material type, EH&S tag, How full

A lot is one record for one physical thing on a shelf: one roll of cloth, one jug of resin or hardener, or one record with a count for a consumable (CS-014 §4). Material type is what kind of thing it is across every container, such as `IN2` or `AT30`.

Also called: lot, batch, jug, roll, container, kit; material type; vendor lot, batch number; EH&S tag, UC tag, RSS tag, Data Matrix; expiry, expiration, use-by; How full; Hazard; stock walk; min stock, minimum, Running out.

- Lot IDs: `FAB-SN6-###`, `RSN-SN6-###` (resin and hardener share it; the Role says which) and `CON-SN6-###` (CS-014 §4).
- Vendor lot number: the batch number the manufacturer printed. Blank when none is printed.
- EH&S tag: the UC EH&S barcode sticker campus puts on every chemical container. A Data Matrix square marked RSS with a 24-character code. FEB never prints one (CS-014 §4).
- State: Sealed, Open, Empty or Expired. How full: Full, Half, Low or Empty (CS-014 §4).
- Hazard: flammable, not flammable, or blank, which means unknown (CS-014 §4).
- Expiry from: where the Expires date came from: vendor label, vendor label (AI read), or shelf-life table (CS-014 §4).
- Incoming: bought and not yet fully received (CS-014 §4).
- Min-stock: the level that triggers a reorder. Stock walk: the monthly check of every shelf against its record. Home location: the shelf (`BIN-SN6-###`) a thing belongs on (CS-011 §4).
- AI read: anything a button filled from a photo. A draft until a person checks it (CS-014 §4).

12. Label words: label, class word, QR code, R&D mark, engraving

A label is the printed 4 × 1 in tag with the name, the ID, a key line and a QR code (CS-001 §4). The QR code opens a public page naming the object, from any phone camera, with no account needed.

Also called: label, sticker, tag; QR, QR code, barcode, scan code; class word; R&D mark, R&D box; mark, marker, paint marker; engraving, engraved ID; Data Matrix, EH&S tag.

- Class word: what kind of object a record is (PART, MOLD, WORK ORDER, FABRIC, RESIN and so on). The app derives it from the ID prefix; it is not printed on the label except STORAGE (CS-001 §4).
- R&D mark: the word R&D in a ruled box on the ID line. Not a class word (CS-001 §4).
- Mark: a hand-written ID in oil-based paint marker where a label cannot go (CS-001 §4).
- Engraving: the ID cut into a mold during machining. The record of record for a mold (CS-001 §4).
- Data Matrix: the square code on an EH&S tag. FEB's own labels are QR; the app's scanner reads both (CS-001 §4).
- Custom label: a label with no record. Never for anything with a record (CS-001 §7.13).
- Which label goes where is QR-005.

13. Trim, sand and bond words: trim line, keying, open time, handling strength

The trim line is the part's cut boundary, transferred from the mold's scribe lines at demold. Keying is abrading a bond face so adhesive can grip; the team uses 120 grit (CS-009 §4).

Also called: trim, trimming, trim line, edge; sand, sanding, key, keying, scuff; open time, work life; handling strength, clamp time; load-carrying bond, structural bond, insert, rod end; bond, glue, bonding.

- Open time: the longest a joint may wait after adhesive goes on a face before it is closed and fixed. 20 min for DP420 (CS-009 §4).
- Handling strength: the point where a bonded joint can be unclamped and moved, not loaded. 2.5 h at 23 °C (73 °F) for DP420 (CS-009 §4).
- Load-carrying bond: any bond that carries load on the car: an insert, a rod end, a mount, a bracket (CS-009 §4).
- Sacrificial ply: the 88 spread-tow outer layer that absorbs sanding; it is the sanding budget, and structural plies are not (CS-009 §4; QR-009 §10).
- Grounding point: the fastener or tab where a part's conductive path meets chassis ground. Surface path: a conductor added to a cured part's surface (CS-010 §4).
- Steps are in QR-013.

14. People and roles: Composites Lead, ME, RE, certified operator

The Composites Lead signs each standard's Approver row (CS-000 §4). On each part, the ME (Mold Engineer) and RE (Manufacturing Engineer) are the two engineer fields, picked from the app's roster (CS-002 §4, CS-013 §4).

Also called: lead, comp lead, composites lead; ME, mold engineer; RE, manufacturing engineer; certified operator, HOT; purchaser; approver; Paul; Steve.

- Lead: a lead is not trained by being a lead. A lead may sign a step that needs training without it, and writes why (CS-013 §4).
- Person fields: every field that names a person is picked from the roster. Use "name" only for someone not on the app (CS-013 §4).
- Certified operator: did the RFS ShopSabre training and HOT. RFS's own record is the authority; the app's ShopSabre CNC training mirrors it (CS-005 §4).
- Larry: reviews the CAM you made yourself before it runs; G10 runs only with his say-so (CS-005 §2, §4).
- ESO: confirms HV isolation before anyone measures resistance on an installed part (CS-010 §6).
- Purchaser: a member who fronts money and gets reimbursed (CS-012 §4).
- Approvers by amount: under $50 the subteam lead, $50 to $250 the division lead, over $250 the President or VP (CS-012 §4).
- Stock-walk owner: named by the lead at season start (CS-011 §7.1).
- Paul: the app's helper, which searches these documents. Steve: the team's joke helper; nothing he says is a rule.

15. Places and channels: RFS, Jacobs basement, etch locker, #composites

RFS is Richmond Field Station, where the team manufactures: the ShopSabre, the RFS container and the shop are there. Finished parts live in the Jacobs basement, never the etch locker (CS-011 §7.3).

Also called: RFS, Richmond, the shop, the field station; Jacobs, Jacobs Hall, Jacobs basement; etch locker; RFS container, container; flammables cabinet; General Box; Slack, #composites, #aerodynamics, #purchasing-slackbot, #purchasing; weekly meeting.

- Storage sites as the app names them: RFS container, Jacobs basement, Flammables cabinet, Dry sealed bin, General Box, Other (CS-011 §7.3).
- Etch locker: finished parts are never stored there (2026-04-20 agreement, CS-011 §7.3).
- Jacobs Hall also has the OMAX waterjet (CS-009 §5).
- #composites: the team's channel. #aerodynamics: where aero parts are often defined.
- #purchasing-slackbot: where purchases are requested and reimbursed through the bot. #purchasing was retired on 2026-10-07 (CS-012).
- Weekly meeting: the composites meeting, now on Wednesdays. Older documents say "Monday meeting" (CS-000 Rev D history).
- Which shelf holds what is QR-014.

16. Purchasing words: reference ID, REQ number, /purchase, Charged to

A reference ID is the budget line a purchase is paid from, such as `COMP-011`; composites lines start `COMP-` (CS-012 §4). A REQ number is the purchasing bot's number for a request, such as `REQ-000064`.

Also called: reference ID, budget line, budget code, COMP-, TRAN-012, -000; REQ, request number; purchasing bot, Flux, the bot; /purchase, /bigorder, /reference, /reimburse; Charged to; Order status, Reimbursement; reimburse, reimbursement, paid back.

- Purchasing bot: the Flux bot in #purchasing-slackbot. `/purchase` requests one purchase, `/bigorder` several items on different reference IDs, `/reference` lists the reference IDs, `/reimburse` takes a reference ID and an amount (CS-012 §4).
- A `-000` ID means an item the budget never listed (CS-012 §4).
- Charged to: the Budget tab field naming the budget a purchase lands on. Blank, or Composites, means ours (CS-012 §4).
- Order status (the goods) and Reimbursement (the money): the Budget tab's two tracks for one purchase (CS-012 §4).
- Restock: the Purpose for the shared Easy Composites order, which replaced SN5's "basket" doc (CS-012 §7.4).
- How to buy something and get paid back is QR-015.

17. Machining words: ShopSabre, HOT, ATC, CAM review, Max-Z

The ShopSabre is the CNC router at RFS that cuts the team's molds. HOT is the RFS hands-on-training sign-off you need before you run it on your own (CS-005 §4). The Shopbot in Jacobs is the machine the team used before it.

Also called: ShopSabre, CNC, router, Shopbot, HOT, hands-on training, ATC, tool changer, CAM review, Larry, max-Z, cut depth.

- Max-Z: the machine's maximum cut depth. A taller mold is split into sections with alignment datums unless the split is waived (CS-003 §4).
- ATC: the automatic tool changer. It swaps tools from its rack mid-program and measures each tool's length, so a tool change needs no manual re-zero (CS-005 §4).
- CAM review: Larry reviews the CAM you made yourself before it runs (CS-005 §4).
- Certified operator: did the RFS ShopSabre training and HOT. RFS's record is the authority, and the app's ShopSabre CNC training mirrors it (CS-005 §4).